

## 34.3 The likelihood equations

In the normal linear model the maximum likelihood estimate was available in closed form, because the log-likelihood was a quadratic in  $\boldsymbol{\beta}$  (Theorem 7.3.1). Outside that case there is no closed form, and everything has to be read off the derivatives of the log-likelihood. This section computes them. The answer has the same shape as the normal equations  $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{0}$ , with two modifications: each observation is weighted according to its variance, and the residual  $y_i - \mu_i$  is converted from the mean scale to the linear-predictor scale.

Throughout, the model is Definition 34.2.1,  $\text{rank}(\mathbf{X}) = p$ , and  $\boldsymbol{\beta}$  lies in the interior of the admissible set  $\mathcal{B}$ . Write

$$\mu_i = h(\eta_i), \quad \eta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}, \quad \frac{d\mu_i}{d\eta_i} = h'(\eta_i),$$

and note that  $h' > 0$  everywhere, because  $g$  is strictly monotone (take  $g$  increasing; a decreasing link only flips signs).

### 34.3.1 The score

#### Theorem 34.3.1. The likelihood equations and the information

Define the diagonal matrices

$$\mathbf{W} = \text{diag}\left\{\frac{w_i h'(\eta_i)^2}{V(\mu_i)}\right\}, \quad \mathbf{D} = \text{diag}\{h'(\eta_i)\}.$$

Then, with  $\ell(\boldsymbol{\beta})$  the log-likelihood of Eq. 34.1.3:

(a) the score is

$$\mathbf{U}(\boldsymbol{\beta}) = \frac{\partial \ell}{\partial \boldsymbol{\beta}} = \frac{1}{\phi} \mathbf{X}^\top \mathbf{W} \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu}), \quad (34.3.1)$$

so the likelihood equations are  $\sum_i x_{ij} w_i h'(\eta_i) \{y_i - \mu_i\} / V(\mu_i) = 0$  for  $j = 1, \dots, p$ , and they do not involve  $\phi$ ;

(b)  $\mathbf{E} \mathbf{U} = \mathbf{0}$  and the **expected (Fisher) information** is

$$\mathcal{I}(\boldsymbol{\beta}) = \text{Cov}\{\mathbf{U}(\boldsymbol{\beta})\} = \frac{1}{\phi} \mathbf{X}^\top \mathbf{W} \mathbf{X}; \quad (34.3.2)$$

(c) writing  $a(\eta) = h'(\eta)/V\{h(\eta)\}$ , the **observed information** is

$$-\frac{\partial^2 \ell}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}^\top} = \frac{1}{\phi} \mathbf{X}^\top (\mathbf{W} - \mathbf{D}_r) \mathbf{X},$$

where  $\mathbf{D}_r = \text{diag}\{w_i(y_i - \mu_i) a'(\eta_i)\}$ ;

(d) under the canonical link  $\mathbf{D}_r = \mathbf{0}$ , so the observed and expected information coincide,  $\mathbf{W} = \text{diag}\{w_i V(\mu_i)\}$ , and Eq. 34.3.1 becomes

$$\mathbf{U}(\boldsymbol{\beta}) = \frac{1}{\phi} \mathbf{X}^\top \text{diag}(\mathbf{w})(\mathbf{y} - \boldsymbol{\mu}). \quad (34.3.3)$$

*Proof.* By Eq. 34.1.3,  $\ell = \phi^{-1} \sum_i w_i \{y_i \theta_i - b(\theta_i)\}$  plus a term free of  $\boldsymbol{\beta}$ . The chain rule gives

$$\frac{\partial \theta_i}{\partial \beta_j} = \frac{d \theta_i}{d \mu_i} \frac{d \mu_i}{d \eta_i} \frac{\partial \eta_i}{\partial \beta_j} = \frac{h'(\eta_i)}{V(\mu_i)} x_{ij},$$

using  $d\theta/d\mu = 1/V(\mu)$  from Definition 34.1.2. Since  $b'(\theta_i) = \mu_i$ ,

$$\frac{\partial \ell}{\partial \beta_j} = \frac{1}{\phi} \sum_i w_i \{y_i - b'(\theta_i)\} \frac{\partial \theta_i}{\partial \beta_j} = \frac{1}{\phi} \sum_i x_{ij} \frac{w_i h'(\eta_i)}{V(\mu_i)} (y_i - \mu_i),$$

which is (a), since  $\mathbf{W}\mathbf{D}^{-1} = \text{diag}\{w_i h'(\eta_i)/V(\mu_i)\}$ . The dispersion appears only as the overall factor  $\phi^{-1}$ , so setting the score to zero gives equations free of  $\phi$ .

- (b)  $E(Y_i - \mu_i) = 0$  gives  $E\mathbf{U} = \mathbf{0}$ . The  $Y_i$  are independent with  $\text{Var}(Y_i) = \phi V(\mu_i)/w_i$  by Proposition 34.1.2, so with the scalars  $a_i = w_i h'(\eta_i)/V(\mu_i)$ ,

$$\text{Cov}(\mathbf{U}) = \frac{1}{\phi^2} \sum_i a_i^2 \text{Var}(Y_i) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top = \frac{1}{\phi} \sum_i \frac{w_i h'(\eta_i)^2}{V(\mu_i)} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top,$$

which is Eq. 34.3.2.

- (c) With  $a(\eta) = h'(\eta)/V\{h(\eta)\}$  we have  $\partial \ell / \partial \beta_j = \phi^{-1} \sum_i x_{ij} w_i (y_i - \mu_i) a(\eta_i)$ . Differentiating again, and using  $\partial \mu_i / \partial \beta_k = h'(\eta_i) x_{ik}$ ,

$$\frac{\partial^2 \ell}{\partial \beta_j \partial \beta_k} = \frac{1}{\phi} \sum_i x_{ij} x_{ik} w_i \{-h'(\eta_i) a(\eta_i) + (y_i - \mu_i) a'(\eta_i)\}.$$

Now  $w_i h'(\eta_i) a(\eta_i) = w_i h'(\eta_i)^2 / V(\mu_i)$  is the  $i$ th working weight, which gives the stated form. Taking expectations kills the second term and recovers (b), a second proof of Eq. 34.3.2.

- (d) Under the canonical link  $\theta_i = \eta_i$ , so  $h = b'$  and  $h'(\eta) = b''(\theta) = V(\mu)$ . Hence  $a(\eta) \equiv 1$ ,  $a' \equiv 0$ ,  $\mathbf{D}_r = \mathbf{0}$ , and  $\mathbf{W} = \text{diag}\{w_i V(\mu_i)^2 / V(\mu_i)\} = \text{diag}\{w_i V(\mu_i)\}$ . Also  $\mathbf{W}\mathbf{D}^{-1} = \text{diag}\{w_i V(\mu_i) / V(\mu_i)\} = \text{diag}(\mathbf{w})$ , which is Eq. 34.3.3. ■

For the normal family with the identity link and  $w_i = 1$ ,  $h' \equiv 1$  and  $V \equiv 1$ , so Eq. 34.3.1 reads  $\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) / \sigma^2$  and Eq. 34.3.2 reads  $\mathbf{X}^\top \mathbf{X} / \sigma^2$ : the score is the normal equations divided by the variance, and Proposition 7.4.7 is recovered.

Three features of Eq. 34.3.1 are worth naming. The equations are **weighted** by  $w_i h'(\eta_i) / V(\mu_i)$ , so a more variable observation counts for less — the logic of weighted least squares (Theorem 21.3.1(a)), with weights dictated by the family rather than assumed. They are **free of**  $\phi$ , which multiplies the whole score, so  $\phi$  can be estimated afterwards without disturbing  $\hat{\boldsymbol{\beta}}$  (Section 34.5), just as least squares does not need  $\sigma^2$ . And they involve the family **only through**  $V$ : two densities with the same variance function give the same  $\hat{\boldsymbol{\beta}}$ , the seed of the quasi-likelihood of Chapter 38 (Definition 38.2.1).

## 34.3.2 The fitted marginals

**Corollary 34.3.2. Canonical links reproduce weighted column totals**

Under the canonical link, any solution  $\hat{\boldsymbol{\beta}}$  of the likelihood equations satisfies

$$\sum_{i=1}^n w_i x_{ij} \hat{\mu}_i = \sum_{i=1}^n w_i x_{ij} y_i, \quad j = 1, \dots, p.$$

In particular, if the model contains an intercept,  $\sum_i w_i \hat{\mu}_i = \sum_i w_i y_i$ ; and if a column of  $\mathbf{X}$  is the indicator of a group, the fitted total over that group equals the observed total.

*Proof.* Immediate from [Eq. 34.3.3](#): the likelihood equations are  $\mathbf{X}^\top \text{diag}(\mathbf{w})(\mathbf{y} - \hat{\boldsymbol{\mu}}) = \mathbf{0}$ , which is the statement column by column. ■

This is the exact analogue of the least squares identity  $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$  of Theorem 6.4.1, and sharper than it looks: not merely that the residuals average to zero, but that a canonical-link fit *reproduces a table of observed totals exactly*, on the original scale of the data, whatever else is in the model.

**Example. Fitted votes by party identification**

The 1996 American National Election Studies subset records, for 944 respondents, a reported vote (coded 1 for Dole), a party-identification score from 0 (strong Democrat) to 6 (strong Republican), and age. Fit a logistic model with one indicator column per party-identification level and age (in decades, centred at 45) as an eighth column. The age coefficient is 0.1044 with standard error 0.0730.

By [Corollary 34.3.2](#) the fitted probabilities must add up, within each of the seven party-identification groups, to the observed number of Dole voters — though age is in the model and varies within every group. Among the 200 strong Democrats, 3 reported voting for Dole and the fitted probabilities sum to 3 exactly; among the 175 strong Republicans observed and fitted totals are both 167. The eighth column gives  $\sum_i \text{age}_i y_i = \sum_i \text{age}_i \hat{\mu}_i = 121.3000$ .

A probit refit destroys the identity: the group totals are then wrong by up to 0.0966 and the age total by 0.6245. The discrepancies are small, because the fits are similar, but nothing forces them to be zero. The same contrast holds for the information, which under the logit link agrees with the expected information to machine precision and under the probit link differs by a relative 0.0056.

```
import numpy as np
import statsmodels.api as sm

anes = sm.datasets.anes96.load_pandas().data
y = anes["vote"].to_numpy(float)           # 1 if the respondent voted
                                           # for Dole
pid = anes["PID"].to_numpy(int)            # party identification, 0 (
                                           # Dem) to 6 (Rep)
D = np.eye(7)[pid]                         # one indicator column per
                                           # level
age = (anes["age"].to_numpy(float) - 45.0) / 10.0
X = np.column_stack([D, age])              # no separate intercept: D
```

```

already sums to 1

logit = sm.GLM(y, X, family=sm.families.Binomial()).fit()
mu = logit.fittedvalues

print("level    n    observed Dole votes    fitted")
for k in range(7):
    inside = pid == k
    print(f"    {k}    {inside.sum():4d}    {y[inside].sum():8.0f}    {mu[inside].sum():"
          f":12.6f}")
print(f"age column: X^T y = {age @ y:.6f},    X^T mu = {age @ mu:.6f}")

```

[Run this code online](#)

*Remark* (Where the marginal property is used). Chapter 37 turns Corollary 34.3.2 into a tool: for a log-linear model of a contingency table the canonical link forces the fitted counts to match the observed margins of the terms in the model, which is what makes log-linear and multinomial fits agree (Theorem 37.3.1). In the binary case it means those totals carry no information about goodness of fit — a point Proposition 35.4.1 takes up.

### 34.3.3 Concavity, existence and uniqueness

#### Theorem 34.3.3. The canonical-link log-likelihood is concave

Under the canonical link,  $\ell(\boldsymbol{\beta})$  is concave on the convex set  $\mathcal{B}$ . If in addition  $\text{rank}(\mathbf{X}) = p$  and  $V(\mu_i) > 0$  for all  $i$ , then  $\ell$  is strictly concave, so it has at most one stationary point, and any stationary point is the unique global maximizer.

*Proof.* Under the canonical link  $\theta_i = \eta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , so by Eq. 34.1.3

$$\ell(\boldsymbol{\beta}) = \frac{1}{\phi} \sum_{i=1}^n w_i \{y_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta} - b(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})\} + \text{const.}$$

The first term is linear in  $\boldsymbol{\beta}$ . In the second,  $b$  is convex by Lemma 34.1.1 and  $\boldsymbol{\beta} \mapsto \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  is linear, so  $\boldsymbol{\beta} \mapsto b(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})$  is convex; multiplying by  $w_i/\phi > 0$  and summing preserves convexity, and the minus sign makes  $\ell$  concave. The set  $\mathcal{B}$  is convex, being an intersection of the preimages of the interval  $g(\mathcal{M})$  under linear maps.

For strictness, Theorem 34.3.1(d) gives the Hessian  $-\phi^{-1} \mathbf{X}^\top \text{diag}\{w_i V(\mu_i)\} \mathbf{X}$ , which for  $\mathbf{v} \neq \mathbf{0}$  satisfies

$$\mathbf{v}^\top \left( \frac{1}{\phi} \mathbf{X}^\top \text{diag}\{w_i V(\mu_i)\} \mathbf{X} \right) \mathbf{v} = \frac{1}{\phi} \sum_i w_i V(\mu_i) (\mathbf{x}_{(i)}^\top \mathbf{v})^2 > 0,$$

because  $\text{rank}(\mathbf{X}) = p$  forces  $\mathbf{X}\mathbf{v} \neq \mathbf{0}$ . So the Hessian is negative definite everywhere on  $\mathcal{B}$  and  $\ell$  is strictly concave there. ■

Concavity settles uniqueness but not existence: a strictly concave function on an open set need not attain its supremum. Both failures happen in practice.

**Example. Two ways the maximum fails to exist**

*All counts zero.* For a Poisson log-linear model with an intercept, if every  $y_i = 0$  then  $\ell(\boldsymbol{\beta}) = -\sum_i e^{\mathbf{x}_{(i)}^\top \boldsymbol{\beta}}$ , which increases to its supremum 0 as  $\beta_0 \rightarrow -\infty$  and is never attained.

*Separated binary data.* If there is a vector  $\mathbf{v}$  with  $\mathbf{x}_{(i)}^\top \mathbf{v} > 0$  whenever  $y_i = 1$  and  $\mathbf{x}_{(i)}^\top \mathbf{v} < 0$  whenever  $y_i = 0$ , then along the ray  $t\mathbf{v}$  every fitted probability moves towards the observed response, so  $\ell$  increases without reaching a maximum. Chapter 35 proves that this is the only way the binary maximum likelihood estimate can fail to exist (Theorem 35.3.1).

The general statement belongs to exponential-family theory. Under a canonical link  $\mathbf{T} = \mathbf{X}^\top \text{diag}(\mathbf{w})\mathbf{y}$  is the sufficient statistic of Section 34.1, and by Corollary 34.3.2 the likelihood equations say  $\mathbf{X}^\top \text{diag}(\mathbf{w})\hat{\boldsymbol{\mu}} = \mathbf{T}$ . The attainable mean-value vectors  $\{\mathbf{X}^\top \text{diag}(\mathbf{w})\boldsymbol{\mu} : \boldsymbol{\mu} \in \mathcal{M}^n\}$  form an open convex set, and the maximum likelihood estimate exists precisely when  $\mathbf{T}$  lies inside it rather than on its boundary (Barndorff-Nielsen 1978, chapter 9; not reproduced here). Both examples above are boundary cases: all-zero counts put  $\mathbf{T}$  at the corner  $\mathbf{0}$ , separation puts it on a face. For non-canonical links even uniqueness can fail.

*Warning.* Concavity is a property of the canonical link, not of generalized linear models. Take the gamma family with the identity link, where  $\ell = \phi^{-1} \sum_i \{-y_i/\mu_i - \log \mu_i\}$  with  $\mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ . The second derivative of one term with respect to  $\mu_i$  is  $(\mu_i - 2y_i)/\mu_i^3$ , positive whenever  $\mu_i > 2y_i$ , so the Hessian is a difference of nonnegative definite matrices and can be indefinite: an iteration can stop at a local maximum. Wedderburn (1976) settles existence and uniqueness link by link, cleanly for the binomial and Poisson with the standard links and not for the gamma with the identity link.

## 34.3.4 Exercises

### 34.3.4.1 A. Check your understanding

**Exercise. A1**

Write the likelihood equations for a Poisson log-linear model with an intercept. Show that they are  $\mathbf{X}^\top (\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$ , that  $\sum_i \hat{\mu}_i = \sum_i y_i$ , and that for the intercept-only model  $\hat{\mu}_i = \bar{y}$ . (This is Exercise (B2), now a special case of Corollary 34.3.2.)

### 34.3.4.2 B. Practice

**Exercise. B1**

For grouped binomial data with  $y_i = s_i/m_i$ , the logit link and  $w_i = m_i$ , write the likelihood equations in terms of the observed successes  $s_i$ . Show that the working weight is  $\mathbf{W}_{ii} = m_i \mu_i (1 - \mu_i)$  and that the information is  $\mathbf{X}^\top \text{diag}\{m_i \mu_i (1 - \mu_i)\} \mathbf{X}$ . Where have you seen that matrix before?

*Solution.* The canonical form Eq. 34.3.3 with  $w_i = m_i$  and  $\phi = 1$  gives  $\sum_i x_{ij} m_i (s_i/m_i - \mu_i) = \sum_i x_{ij} (s_i - m_i \mu_i) = 0$ : fitted and observed successes agree in every column of  $\mathbf{X}$ . By Theorem 34.3.1(d),  $\mathbf{W} = \text{diag}\{w_i V(\mu_i)\}$  with  $V(\mu) = \mu(1 - \mu)$ , giving  $\mathbf{W}_{ii} = m_i \mu_i (1 - \mu_i)$  and the stated information. It is the weight matrix of the empirical-logit weighted least squares fit in Exercise (B2), with  $\mu_i$  in place of  $\hat{p}_i$ : that procedure is one step of the iteration of Section 34.4 started from the data.

### Exercise. B2

Show that the observed information equals the expected information *identically in  $\mathbf{y}$*  if and only if the link is canonical (up to an affine change of  $\eta$ ). Assume that no row of  $\mathbf{X}$  is zero, an observation with  $\mathbf{x}_{(i)} = \mathbf{0}$  carrying no information about  $\boldsymbol{\beta}$ . Use Theorem 34.3.1(c).

*Solution.* By Theorem 34.3.1(c) the two agree for all  $\mathbf{y}$  iff  $\mathbf{X}^\top \mathbf{D}_r \mathbf{X} = \mathbf{0}$  for all  $\mathbf{y}$ , where  $(\mathbf{D}_r)_{ii} = w_i (y_i - \mu_i) a'(\eta_i)$ . Vary  $y_i$  over two admissible values with the other responses held fixed: subtracting the two identities leaves  $\delta_i a'(\eta_i) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top = \mathbf{0}$  with  $\delta_i = w_i (y_i - y'_i) \neq 0$ , and since  $\mathbf{x}_{(i)} \neq \mathbf{0}$  this forces  $a'(\eta_i) = 0$  for every  $i$ . (Two admissible values always exist: the  $y_i$  range over an interval for the normal and gamma families, and over  $\{0, 1/m, \dots, 1\}$  or  $\{0, 1, 2, \dots\}$  for the binomial and Poisson.) So  $a$  is constant, say  $a \equiv 1/\lambda$ , that is  $d\mu/d\eta = V(\mu)/\lambda$ . Comparing with the canonical link, for which  $d\mu/d\theta = V(\mu)$ , gives  $\eta = \lambda\theta + \text{const.}$

### 34.3.4.3 C. Going deeper

#### Exercise. C1

For a binary response,  $\ell(\boldsymbol{\beta}) = \sum_i \{y_i \log h(\eta_i) + (1 - y_i) \log(1 - h(\eta_i))\}$ . Show that  $\ell$  is concave for every design and every data set if and only if  $\log h$  and  $\log(1 - h)$  are both concave functions of  $\eta$ . Verify that the logit link satisfies this, and that  $h(\eta) = \Phi(\eta)$  does too, using the fact that the standard normal density is log-concave.

#### Exercise. C2

An **offset** is a known column added to the linear predictor with coefficient fixed at 1:  $\eta_i = o_i + \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ . Show that Theorem 34.3.1 holds verbatim with  $\eta_i$  redefined this way, that Corollary 34.3.2 is unchanged, and that the offset does *not* appear among the columns whose totals are reproduced. Where would such a term come from in a Poisson model for rates? (Chapter 37 answers this with Proposition 37.2.1.)